

Subject Description Form

Subject Code	EEE507
Subject Title	Quantum Microdevices
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject provides comprehensive study of the principle, theories and applications of quantum microdevice devices. Understanding the fundamental design and fabrication of quantum microdevices, and know quantum microdevice control and experiment characterization. Application of single-photon devices, quantum switches, single-photon detectors and quantum sensing in optical, magnetic and gravity etc.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills - Understand the fundamentals physics and basic principles concept of quantum microdevice - Awareness of quantum device design, simulation and fabrication - Understand control actions, measurements, and decoherence properties of quantum sensing in optical, magnetic and gravity applications. Category B: Attributes for all-roundedness - Communicate effectively. - Think critically and creatively. - Assimilate new technological development in related field
Subject Synopsis/ Indicative Syllabus	Overview of quantum devices and microfabrication explain the basic concept, fundamentals theory and formalisms of quantum devices and quantum systems. Physics of quantum microdevice the fundamental concept and principle of different of promising quantum devices, namely cavity QED, ion traps, neutral atom traps, quantum dots, linear optics, superconducting quantum interference devices (SQUID) etc. Design and simulation of the quantum microdevices, understanding design tool and case studies of single-photon devices, quantum switches and single-photon detectors. Microfabrication technology with different materials properties and processes are introduced: Silicon SOI wafer fabrication, Si_3N_4 fabrication process and Lithium Niobate fabrication process. Quantum sensors for each device, the physical properties of the quantum sensor; the control actions, measurements, and decoherence properties of the device; quantum sensing in optical, magnetic and gravity applications. Linear optics devices light beam manipulation with four degrees of freedom, optical circuits and examples; quantum optical states; quantum operations and gates: Knill, Laflamme and Milburn; and quantum teleportation.

Teaching/Learning Methodology	The basic concepts of micromachined systems will be introduced in lectures. The student will understand the micromachined systems design flow and main components of optoelectronic systems. Students will also be required to understand the fundamentals of quantum key distribution systems, and optical detection magnetic resonance in lectures.																																												
	Teaching/Learning Methodology		Intended subject learning outcomes																																										
		a	b	c	d	e	f																																						
Lectures		✓	✓	✓	✓	✓																																							
Assignments & Report		✓	✓	✓	✓	✓	✓																																						
Examinations		✓	✓	✓																																									
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>1. Assignment & Report</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments: Assignments are given to students to assess their competence level of knowledge and comprehension. The criteria (i.e. what to be demonstrated) and level (i.e. the extent) of achievement will be graded. Report: Students will be required to submit a report, by reviewing the material taught in the lectures as well as their self-study and readings. End-of-semester Examination: There will be an end-of-semester examination to assess students’ achievement of all the learning outcomes							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Assignment & Report	40%	✓	✓	✓	✓	✓	✓	2. Examination	60%	✓	✓	✓				Total	100 %						
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Total	100 %																																												
Student Study Effort Expected	Class contact:																																												
	▪ Lectures					30 Hrs.																																							
	▪ Tutorials					9 Hrs.																																							
	Other student study effort:																																												
	▪ Reading and self-study					60 Hrs.																																							
	▪ Assignment & Report					21 Hrs.																																							
	Total student study effort					120 Hrs.																																							

Reading List and References	Textbooks 1. G. Chen, David A. Church, B. G. Englert, C. Henkel, B. Rohwedder, M. O. Scully and M. S. Zubairy, Quantum Computing Devices Principles, Design and Analysis, Taylor & Francis Group, LLC, 2007. References 1. Manijeh Razeghi, Technology of Quantum Devices, Springer 2010. 2. Quantum Materials, Devices, and Applications, Henini, Mohamed; Rodrigues, Marcelo Oliveira, United States: Elsevier, 2022 3. J. A. Del Alamo, Integrated Microelectronic Devices: Physics and Modeling, Pearson, 2017 4. Integrated microelectronic devices : physics and modeling, Alamo, Jesús A. del (author), New York : Pearson; 2018
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